TASK 4

Implement a Map Reduce program to mine and analyze weather data set.

mm.py

```
#!/usr/bin/env python
import sys
for line in sys.stdin:
    line = line.strip()
    values = line.split()
        if len(values) < 3:
        continue
    try:
        date = values[0]
        max_temp = float(values[1])
        min_temp = float(values[2])
        print "%s\t%.2f\t%.2f" % (date, max_temp, min_temp)
    except ValueError:
        continue
```

rr.py

```
#!/usr/bin/env python
import sys
hot_day = None
cold_day = None
hot_temp = float('-inf')
cold_temp = float('inf')
for line in sys.stdin:
    values = line.strip().split("\t")

    if len(values) != 3:
        continue

    try:
        date = values[0]
        max_temp = float(values[1])
        min_temp = float(values[2])


        if max_temp > hot_temp:
            hot_temp = max_temp
            hot_day = date


        if min_temp < cold_temp:
            cold_temp = min_temp
            cold_day = date

    except ValueError:
        continue
```

```
if hot_day and cold_day:
    print "%s\t%s" % ("Hottest Day:", hot_day)
    print "%s\t%s" % ("Coldest Day:", cold_day)
```

input file (we.txt)

```
20200101	18.8	21.8
20200102	19.1	23.4
20200103	18.4	26.8
20200104	22.0	28.0
20200105	17.5	20.5
20200106	16.2	21.0
20200107	20.0	27.3
20200108	19.8	25.0
20200109	18.0	22.7
20200110	21.5	29.4
```

check mapper output:

hdfs dfs -cat weather/we.txt | python mm.py

check mapper output:

hdfs dfs -cat weather/we.txt | python mm.py | sort | python rr.py

Command to run the output:

hadoop jar /usr/lib/hadoop-mapreduce/hadoop-streaming.jar -input weather/we.txt -output weather/output -mapper "python mm.py" -reducer "python rr.py" -file mm.py -file rr.py